

Project Design Phase-II

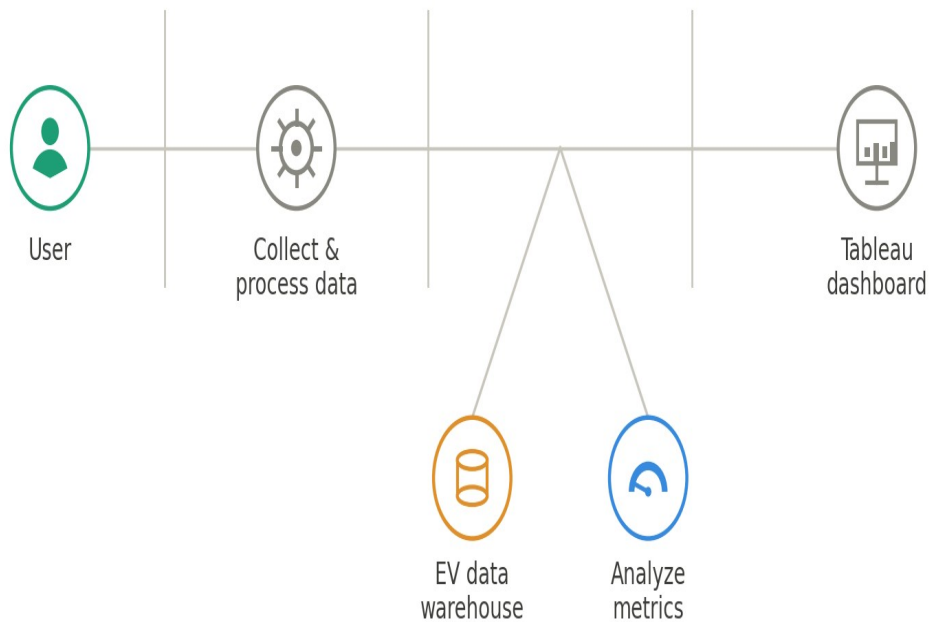
Data Flow Diagram & User Stories

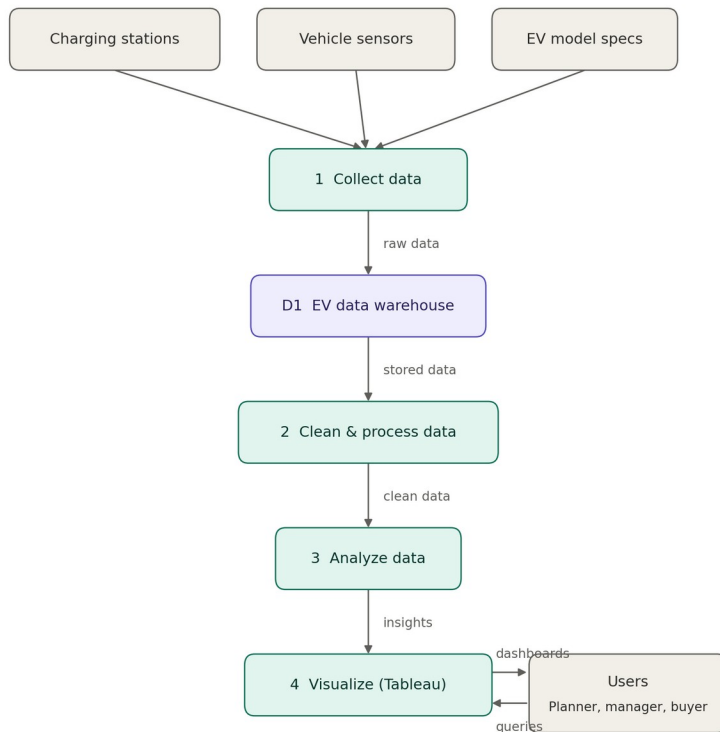
Date	2 July 2026
Team ID	
Project Name	EV Analytics & Visualization Dashboard
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Flow





User Stories

User stories for the EV analytics and visualization dashboard, covering the city energy planner, fleet manager, and consumer personas.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
City energy planner	Charging pattern analysis	USN-1	As a city energy planner, I can view dashboards of charging time, duration, and power consumed across the city so I can understand EV charging demand.	Dashboard shows charging trends by time of day and location	High	Sprint-1
City energy planner	Peak demand visualization	USN-2	As a city energy planner, I can view heatmaps of neighborhoods with high EV charging demand so I can plan new charging infrastructure.	Heatmap highlights high-demand areas by neighborhood	High	Sprint-1
Fleet manager	Battery performance monitoring	USN-3	As a fleet manager, I can track state of charge, depth of discharge, and efficiency for each vehicle so I can monitor battery health.	Dashboard displays battery health metrics per vehicle	High	Sprint-1
Fleet	Range	USN-4	As a fleet manager, I can	Dashboard shows	Medium	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
manager	prediction		compare predicted versus actual driving range so I can spot underperforming vehicles.	predicted vs actual range side by side	um	nt-2
Fleet manager	Maintenance alerts	USN-5	As a fleet manager, I can receive an alert when a vehicle's battery is likely to need replacement so I can reduce downtime.	Alert is triggered when battery health falls below a set threshold	Medium	Sprint-2
Consumer (EV buyer)	Model comparison	USN-6	As a consumer, I can compare different EV models on price, range, and efficiency so I can choose the best vehicle for my needs.	Dashboard shows a side-by-side comparison of multiple EV models	High	Sprint-1
Consumer (EV buyer)	Cost & emissions estimate	USN-7	As a consumer, I can view cost per kilometer and estimated long-term savings for each EV model so I can make an informed financial decision.	Dashboard displays cost and emissions estimates per model	Medium	Sprint-2
Consumer (EV buyer)	Personalized filtering	USN-8	As a consumer, I can filter EV model comparisons by driving style and region so I get personalized recommendations.	Filters update the comparison dashboard by driving style and region	Low	Sprint-2